

Solutions to Problem Solving Class 6: Routing

Computer Communications – DE
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ht 2006

1. The updated routing table is:

Net 2 6 A
Net 3 5 C
Net 4 3 A
Net 6 3 C
Net 7 3 B

2. Minimum hop count from A to E is 3

- 1: A sends 2 packets (B & C).
- 2: B sends 2 packets (C & D), C sends 2 packets (B & D).
- 3: C sends 2 packets (A & D), B sends 2 packets (A & D), D sends 2 packets (C & E), D sends 2 packets (B & E).

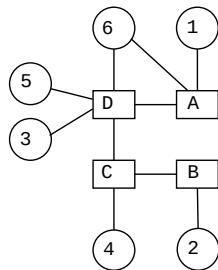
Total amount of packets = $2 \cdot 7 = 14$.

- 3.

Since we are using a Link State protocol, we need to apply Dijkstra's algorithm to solve the shortest path problem:

Iter	M	B	C	D	E	F
1	A	2 AB	1 AC	∞ -	∞ -	∞ -
2	AC	2 AB	1 AC	4 ACD	3 ACE	6 ACF
3	ACB	2 AB	1 AC	4 ACD	3 ACE	6 ACF
4	ACBE	2 AB	1 AC	4 ACD	3 ACE	6 ACF
5	ACBED	2 AB	1 AC	4 ACD	3 ACE	5 ACDF
6	ACBEDF	2 AB	1 AC	4 ACD	3 ACE	5 ACDF

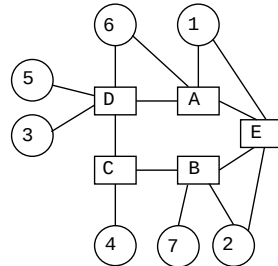
4. a)



b)

Subnet	Cost	Next-hop
1	1	-
2	2	E
3	2	D
4	3	D
5	2	D
6	1	-
7	3	E

c)



5. ARP request from MAC(A) to MAC(broadcast): Who has IP(DNS)?
 ARP reply from MAC(DNS) to MAC(A): I am IP(DNS)
 DNS request from MAC(A) to MAC(DNS): from IP(A) to IP(DNS): What IP is hostname(D)?
 DNS reply from MAC(DNS) to MAC(A): from IP(DNS) to IP(A): host-name(D) = IP(D)
 ARP request from MAC(A) to MAC(broadcast): Who has IP(R1)?
 ARP reply from MAC(R1) to MAC(A): I am IP(R1)
 ICMP echo from MAC(A) to MAC(R1): from IP(A) to IP(D)
 ICMP redirect from MAC(R1) to MAC(A): from IP(R1) to IP(A); redirect to IP(R2)
 ARP request from MAC(R1) to MAC(broadcast): Who has IP(R2)?
 ICMP reply from MAC(R2) to MAC(A): from IP(D) to IP(A): ICMP echo reply

In the second case the messages are identical (substituting A for B). However, because of the switch there are only three messages seen by the sniffer – the broadcast messages.